

12/8/20

Activity 5

S. THANISHKA
192521457

① [4371, 1323, 6173, 4199, 4344, 9679, 1989]

$$h(x) = x \% 10 \quad \text{S.T.}$$

a) Separate chaining hash Table:

$$h(4371) = 4371 \% 10 = 1$$

$$h(1323) = 1323 \% 10 = 3$$

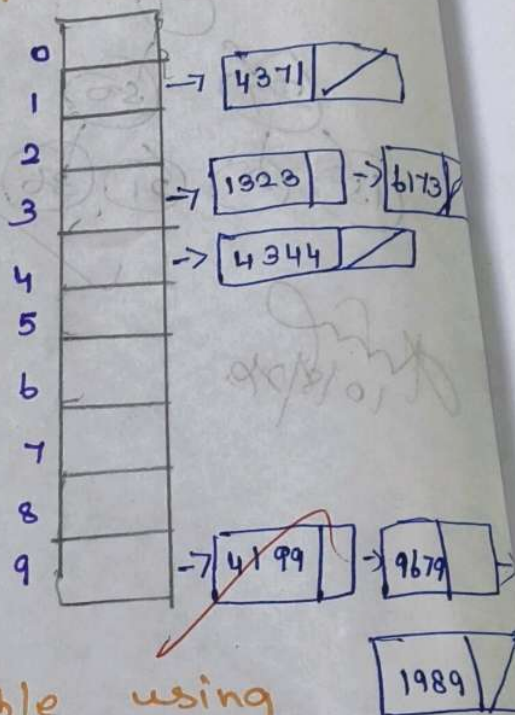
$$h(6173) = 6173 \% 10 = 3$$

$$h(4199) = 4199 \% 10 = 9$$

$$h(4344) = 4344 \% 10 = 4$$

$$h(9679) = 9679 \% 10 = 9$$

$$h(1989) = 1989 \% 10 = 9$$



b) open addressing hash Table using linear probing.

$$h(4371) = 1$$

$$h(1323) = 3$$

$$h(6173) = 3 \quad \text{collision}$$

$$h(6173) = 6173 + 1 = 6174 \quad 6174 \% 10 = 4 \quad \text{probe}$$

$$6173 + 2 = 6175 \quad 6175 \% 10 = 5 \Rightarrow 5$$

$$h(4199) = 9$$

$$h(4344) = 4 \quad 4344 + 1 = 4345 \quad 4345 \% 10 = 5$$

$$h(9679) = 9 \quad \text{collision}$$

$$h(9679) = 9679 + 1 = 9680 \quad 9680 \% 10 = 0$$

$$9679 + 2 = 9681$$

$h(1989) = 9$ collision

$h(1989) = 1989 + 1 = 1990 \div 10 \rightarrow 0$ probe

$h(1989) = 1989 + 2 = 1991 \div 10 \rightarrow 1$ probe = 1

$h(1989) = 1989 + 3 = 1992 \div 10 \rightarrow 2$

0	9679
1	4371
2	1989
3	1323
4	6173
5	4371 4344
6	
7	
8	
9	4199

c) open addressing hash Table using quadratic probing.

$$(h(x) + i^2) \bmod 10$$

$$h(4371) = 1$$

$$h(1323) = 3$$

$$h(6173) = 3 \text{ collision} \rightarrow 6173 + 1^2 = 6174$$

$$\Rightarrow 6173 + 1^2 = 6174 \div 10 = 4$$

$$h(4199) = 9$$

$$h(4344) = 4 \rightarrow \text{collision} \Rightarrow 4344 + 1^2 = 4345 \div 10 = 5$$

$$h(9679) = 9 \rightarrow 9679 + 1^2 = 9680 \div 10 = 0$$

$$h(1989) = 9 \rightarrow 1989 + 1^2 = 1990 \div 10 = 0 \text{ Probe}$$

$$1989 + 2^2 = 1993 \div 10 = 3 \text{ Probe}$$

$$1989 + 3^2 = 1998 \div 10 = 8$$

0	9679
1	4371
2	
3	1323
4	6173
5	4344
6	
7	
8	1989
9	4199

0	
1	
2	
3	
4	
5	
6	
7	
8	
9	

d) open addressing has Table Double Hashing

$$h_1(x) = x \bmod 10$$

$$h_2(x) = 7 - (x \bmod 7)$$

$$\Rightarrow h(4371) = 1$$

$$\Rightarrow h(1323) = 3$$

$$h(6173) = 3 \text{ collision.}$$

$$(6173 \% 10) + (7 - 6173 \% 7) * 1 = 3 + 4$$

$$h(4371) = 1$$

$$h(1323) = 3$$

$$h(6173) = (3 + 1 \times 1) \% 10 = 4$$

$$h(4199) = 9$$

$$h(4344) = (4 + 1 \times 3) \% 10 = 7$$

~~$$(4 + 2 \times 1) \% 10 = 6$$~~

$$h(9679) = (9 + (7-1) \times 1) = 15$$

$$= 9 + 6 = 15 \% 10 = 5$$

$$h(1989) = 9 + (7-1) \times 3 = 27$$

$$= 9 + 6 = 15$$

$$h(1989) = 9 + 18 = 27$$

$$= 27$$

0	
1	4371
2	
3	1323
4	6173
5	9679
6	
7	4344
8	
9	4199

~~1989~~
 ∴ 1989 cannot be inserted